

Prithvish Baidya

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India · Remote-first · Comfortable with US (PT), EU (CET), or async timezones

Infrastructure engineer. Co-authored [ERC-7518](#). Built the transaction backend powering [American Express's NFT program](#). Won [ETHIndia 2024](#) (2,500+ hackers, 36 hours). 4 years shipping production systems: transaction infrastructure, cryptography, and protocol tooling.

EXPERIENCE

[thirdweb](#) | Software Engineer

Jul 2024 – Nov 2025

Remote

- Built [engine-core](#): Rust transaction backend powering thirdweb's API. 100k+ daily transactions across 2,000+ chains. Nonce management is the hard part, and this makes "mint 10,000 NFTs in one call" actually work.
- Powers [Amex Passport](#), issuing digital collectibles to 140M+ cardholders. Zero failed transactions in production.
- Built [twmq](#): Redis-backed job queue with exactly-once delivery, sub-100ms p99 latency. Handles crash recovery and retries for engine-core.
- Designed **Vault**: non-custodial key management using AWS Nitro Enclaves. Keys live in a TEE with cryptographic attestation. Enterprise custody without the custody risk.

[Zoniq](#) | Lead Architect

Jul 2022 – Jun 2024

Remote

- Led team of 4 engineers. Co-authored [ERC-7518](#): Ethereum standard for compliant real-world asset tokenization. Introduces partition-based semi-fungible tokens with dynamic transfer restrictions.
- Built the first compliant RWA tokenization layer on XRPL.
- Architected enterprise platforms: **Heimdall** (web3 operational layer) and **CompliTO** (compliant security token transfers).

[Biconomy](#) | Software Engineer

Nov 2021 – May 2022

Remote

- Rebuilt gasless transaction infrastructure processing 2M+ meta-transactions/month. Enterprise clients included JP Morgan.
- Built the first multichain **ERC-4337** smart contract wallet (hackathon POC → production in 3 months).

Libertas Network | Senior Fullstack Developer

Mar 2021 – Nov 2021

Remote

- Built decentralized censorship-resistant podcasting platform using IPFS, Node.js, and Ethereum. (Company shut down)

PROJECTS

[zkRail](#) | [ETHIndia 2024 Winner](#) | [Demo Video](#)

Dec 2024

Trustless fiat settlement using ZK proofs over banking APIs. Solver network executes real payments; 150% collateral + game-theoretic bonds prevent fraud. Won 1st place overall (2,500+ hackers, \$50k+ prize pool) plus 4 track bounties. Built in 36 hours solo. [Technical writeup](#).

[t2z](#) | 1st Place ECC Track, Zyperpunk 2025 (\$9k total)

2025

Rust library for Zcash shielded transactions. Compiles Halo 2 ZK proving to 3.7MB WASM (no 50MB parameter downloads). Implements ZIP-374 PCZT with external signer support for hardware wallets. Cross-platform via UniFFI.

[Coredrain](#)

2025

Indexer for Hyperliquid's exotic architecture (HyperCore L1 to HyperEVM). Correlates L1 transfers with EVM transaction hashes. Discovered [undocumented dual-hash convention](#) for system transactions by reverse-engineering nanoreth source. Lazy indexing with block interpolation search.

PUBLICATIONS & OPEN SOURCE

[ERC-7518: Dynamic Compliant Interop Security Token](#), Co-author

Review, Ethereum EIPs

[engine-core](#): Production Rust transaction infrastructure for blockchain. Primary author.

2024–2025

[Bun, PR #25514](#): Added S3 Requester Pays support to Bun's native S3 client. Zig, TypeScript.

2025

STACK

Languages / Rust, Go, TypeScript, Solidity

Infrastructure / PostgreSQL, Redis, Kafka, AWS Nitro Enclaves

Cryptography / ZK (Halo 2, Noir), Zcash/Orchard, WASM

Frontend / React, Next.js, Astro, Tanstack, Three.js

RECOGNITION

[ETHIndia 2024](#): 1st place overall (2,500+ hackers) plus Polygon, Coinbase, CDP, Base bounties

Dec 2024

[Zypherpunk 2025](#): 1st place ECC track (\$5k), plus Tachyon, Raybot, Network School bounties (\$9k total)

2025

KVPY-SA Fellowship: National science fellowship by Indian Institute of Science. Top 1% of 100,000+ applicants selected for research mentorship.

2019

NTSE Scholar: National Talent Search Examination. Nationwide exam selecting top 1,000 students from 1.5M+ for merit scholarship.

2019